

WIFI SECURITY ANALYZER

Wireless Security Assessment Report

Interface: wlan0

Mode: demo

Scan Date: 2026-05-06 17:28:15

Networks Found: 8

This report contains sensitive security information. Restrict access to authorised personnel.

Executive Summary

Total Networks	8
Average Risk Score	35.9 / 100
Critical	0
Vulnerable	4
Moderate	2
Secure	2
Open Networks	2
WEP Networks	1
WPS Enabled	3

Network Overview

#	SSID	BSSID	Signal	Enc	WPS	Score	Risk
1	CoffeeShop_Free	AA:BB:CC:11:22:33	-45 dBm	OPEN	No	54	Vulnerable
2	CoffeeShop_Free	FF:EE:DD:33:22:11	-50 dBm	OPEN	No	54	Vulnerable
3	OldOfficeNetwork	AA:BB:CC:11:22:55	-60 dBm	WEP	No	42	Vulnerable
4	Netgear_ABCD	AA:BB:CC:11:22:99	-80 dBm	WPA	Yes	41	Vulnerable
5	TP-Link_2D4E	AA:BB:CC:11:22:44	-55 dBm	WPA2	Yes	36	Moderate
6	JioFiber_Guest	AA:BB:CC:11:22:77	-70 dBm	WPA2	Yes	34	Moderate
7	<hidden>	AA:BB:CC:11:22:88	-75 dBm	WPA2	No	14	Secure
8	HomeNet_5G	AA:BB:CC:11:22:66	-65 dBm	WPA3	No	12	Secure

Detailed Network Analysis

CoffeeShop_Free AA:BB:CC:11:22:33		Score: 54/100 Vulnerable
* Open / Unencrypted Network		Critical
The network broadcasts without encryption or authentication. Traffic can be captured or modified by anyone within radio range.		
Scenario: A nearby attacker can join the AP without credentials and observe DNS, HTTP, and other clear-text traffic. They may also attempt man-in-the-middle attacks against connected clients.		
Recommendation: Enable WPA3 Personal. If unsupported, use WPA2-AES with a strong random passphrase. Avoid transmitting sensitive data on open hotspots without a VPN.		
* Duplicate SSID		High
The SSID 'CoffeeShop_Free' is broadcast by 2 access points (BSSIDs: AA:BB:CC:11:22:33, FF:EE:DD:33:22:11). This may be normal in managed enterprise deployments, but it is suspicious on small or unmanaged networks.		
Scenario: An attacker can clone a legitimate SSID with a stronger signal. Clients configured to auto-connect may associate with the rogue AP.		
Recommendation: Verify each BSSID against a known AP inventory. Remove unapproved devices and use 802.1X or wireless intrusion monitoring for managed environments.		
CoffeeShop_Free FF:EE:DD:33:22:11		Score: 54/100 Vulnerable
* Open / Unencrypted Network		Critical
The network broadcasts without encryption or authentication. Traffic can be captured or modified by anyone within radio range.		
Scenario: A nearby attacker can join the AP without credentials and observe DNS, HTTP, and other clear-text traffic. They may also attempt man-in-the-middle attacks against connected clients.		
Recommendation: Enable WPA3 Personal. If unsupported, use WPA2-AES with a strong random passphrase. Avoid transmitting sensitive data on open hotspots without a VPN.		
* Duplicate SSID		High
The SSID 'CoffeeShop_Free' is broadcast by 2 access points (BSSIDs: FF:EE:DD:33:22:11, AA:BB:CC:11:22:33). This may be normal in managed enterprise deployments, but it is suspicious on small or unmanaged networks.		
Scenario: An attacker can clone a legitimate SSID with a stronger signal. Clients configured to auto-connect may associate with the rogue AP.		
Recommendation: Verify each BSSID against a known AP inventory. Remove unapproved devices and use 802.1X or wireless intrusion monitoring for managed environments.		
OldOfficeNetwork AA:BB:CC:11:22:55		Score: 42/100 Vulnerable
* WEP Encryption (Broken Protocol)		Critical
WEP was deprecated in 2004 and is cryptographically broken. Its RC4-based design allows key recovery from captured traffic.		
Scenario: An attacker can passively capture enough WEP frames and recover the network key with public tooling, then access the network as an authenticated client.		
Recommendation: Replace WEP with WPA3 immediately. Retire devices that only support WEP and avoid sending sensitive data over this network.		
Netgear_ABCD AA:BB:CC:11:22:99		Score: 41/100 Vulnerable
* WPA-TKIP (Deprecated Protocol)		High
WPA-TKIP was a transitional replacement for WEP and is deprecated. It does not meet modern wireless security requirements.		

Scenario: Attackers may exploit protocol weaknesses or capture authentication handshakes for offline password guessing, especially when passphrases are weak or reused.

Recommendation: Disable WPA and TKIP. Use WPA3 Personal or WPA2-AES with a long random passphrase and current AP firmware.

*** Default / Vendor SSID**

High

The SSID 'Netgear_ABCD' matches a vendor or generic default naming pattern (prefix: 'netgear'). Default naming can reveal device type and may indicate other factory defaults remain unchanged.

Scenario: An attacker can infer router vendor or model family, look up common default credentials, and attempt administrative access against the AP.

Recommendation: Use a unique SSID that does not reveal vendor or model details. Change the router admin password and keep firmware updated.

*** WPS Enabled**

High

Wi-Fi Protected Setup is enabled. WPS PIN mode has design weaknesses that reduce the practical authentication search space.

Scenario: An attacker may attempt online WPS PIN guessing or exploit weak vendor PIN generation to recover the wireless passphrase.

Recommendation: Disable WPS. Replace APs that do not allow WPS to be disabled.

TP-Link_2D4E | AA:BB:CC:11:22:44

Score: 36/100 | Moderate

*** WPA2-PSK (Passphrase Dependent)**

Moderate

WPA2-AES is broadly secure when paired with a strong passphrase, but captured handshakes can be attacked offline if passwords are weak.

Scenario: An attacker who captures a 4-way handshake can run offline dictionary attacks. WPS, default passwords, and short passphrases greatly increase compromise risk.

Recommendation: Use at least 16 random characters, disable WPS, enable Management Frame Protection where supported, and plan a WPA3 upgrade.

*** Default / Vendor SSID**

High

The SSID 'TP-Link_2D4E' matches a vendor or generic default naming pattern (prefix: 'tp-link'). Default naming can reveal device type and may indicate other factory defaults remain unchanged.

Scenario: An attacker can infer router vendor or model family, look up common default credentials, and attempt administrative access against the AP.

Recommendation: Use a unique SSID that does not reveal vendor or model details. Change the router admin password and keep firmware updated.

*** WPS Enabled**

High

Wi-Fi Protected Setup is enabled. WPS PIN mode has design weaknesses that reduce the practical authentication search space.

Scenario: An attacker may attempt online WPS PIN guessing or exploit weak vendor PIN generation to recover the wireless passphrase.

Recommendation: Disable WPS. Replace APs that do not allow WPS to be disabled.

JioFiber_Guest | AA:BB:CC:11:22:77

Score: 34/100 | Moderate

*** WPA2-PSK (Passphrase Dependent)**

Moderate

WPA2-AES is broadly secure when paired with a strong passphrase, but captured handshakes can be attacked offline if passwords are weak.

Scenario: An attacker who captures a 4-way handshake can run offline dictionary attacks. WPS, default passwords, and short passphrases greatly increase compromise risk.

Recommendation: Use at least 16 random characters, disable WPS, enable Management Frame Protection where supported, and plan a WPA3 upgrade.

* Default / Vendor SSID	High
The SSID 'JioFiber_Guest' matches a vendor or generic default naming pattern (prefix: 'jiofiber'). Default naming can reveal device type and may indicate other factory defaults remain unchanged.	
Scenario: An attacker can infer router vendor or model family, look up common default credentials, and attempt administrative access against the AP.	
Recommendation: Use a unique SSID that does not reveal vendor or model details. Change the router admin password and keep firmware updated.	

* WPS Enabled	High
Wi-Fi Protected Setup is enabled. WPS PIN mode has design weaknesses that reduce the practical authentication search space.	
Scenario: An attacker may attempt online WPS PIN guessing or exploit weak vendor PIN generation to recover the wireless passphrase.	
Recommendation: Disable WPS. Replace APs that do not allow WPS to be disabled.	

<hidden> AA:BB:CC:11:22:88	Score: 14/100 Secure
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* WPA2-PSK (Passphrase Dependent)	Moderate
WPA2-AES is broadly secure when paired with a strong passphrase, but captured handshakes can be attacked offline if passwords are weak.	
Scenario: An attacker who captures a 4-way handshake can run offline dictionary attacks. WPS, default passwords, and short passphrases greatly increase compromise risk.	
Recommendation: Use at least 16 random characters, disable WPS, enable Management Frame Protection where supported, and plan a WPA3 upgrade.	

* Hidden SSID	Low
The network suppresses its SSID in beacon frames. This offers little practical protection and can increase client probing privacy leakage.	
Scenario: A passive observer can still identify the hidden network when clients connect or probe for it.	
Recommendation: Do not rely on hidden SSIDs as a security control. Use WPA3 or WPA2-AES with a strong passphrase.	

HomeNet_5G AA:BB:CC:11:22:66	Score: 12/100 Secure
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* WPA3 (Secure Configuration Detected)	Low
WPA3 uses SAE to resist offline dictionary attacks and is the current best practice for personal wireless networks.	
Scenario: WPA3 materially raises attack cost, but outdated firmware or transition mode can still expose downgrade or implementation risks.	
Recommendation: Keep AP firmware current, prefer WPA3-only mode where compatible, and require Management Frame Protection.	

* Default / Vendor SSID	High
The SSID 'HomeNet_5G' matches a vendor or generic default naming pattern (prefix: 'home'). Default naming can reveal device type and may indicate other factory defaults remain unchanged.	
Scenario: An attacker can infer router vendor or model family, look up common default credentials, and attempt administrative access against the AP.	
Recommendation: Use a unique SSID that does not reveal vendor or model details. Change the router admin password and keep firmware updated.	

Consolidated Recommendations

1. Enable WPA3 Personal. If unsupported, use WPA2-AES with a strong random passphrase. Avoid transmitting sensitive data on open hotspots without a VPN.
2. Verify each BSSID against a known AP inventory. Remove unapproved devices and use 802.1X or wireless intrusion monitoring for managed environments.
3. Replace WEP with WPA3 immediately. Retire devices that only support WEP and avoid sending sensitive data over this network.
4. Disable WPA and TKIP. Use WPA3 Personal or WPA2-AES with a long random passphrase and current AP firmware.
5. Use a unique SSID that does not reveal vendor or model details. Change the router admin password and keep firmware updated.
6. Disable WPS. Replace APs that do not allow WPS to be disabled.
7. Use at least 16 random characters, disable WPS, enable Management Frame Protection where supported, and plan a WPA3 upgrade.
8. Do not rely on hidden SSIDs as a security control. Use WPA3 or WPA2-AES with a strong passphrase.
9. Keep AP firmware current, prefer WPA3-only mode where compatible, and require Management Frame Protection.